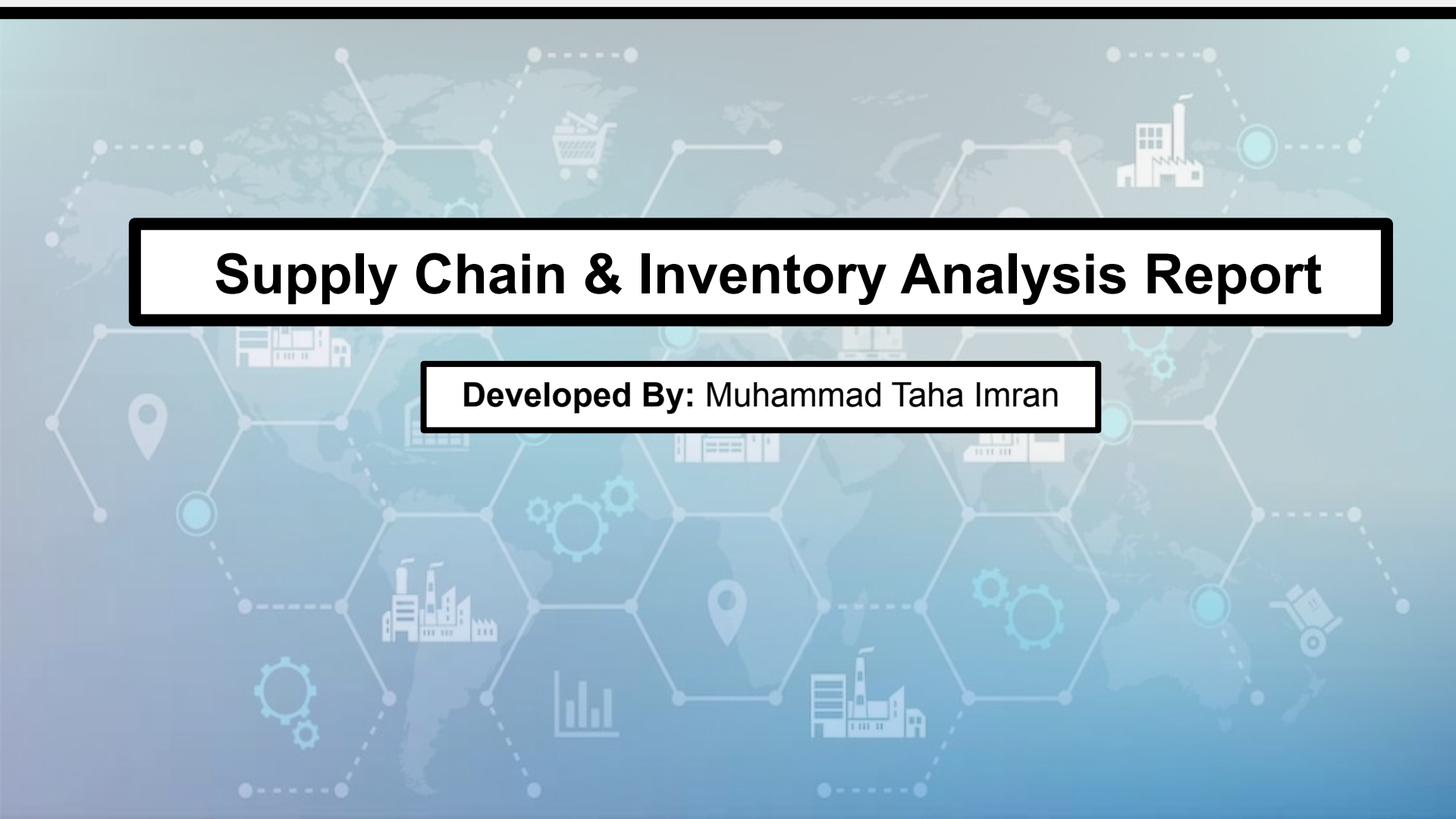




Supply Chain & Inventory Analysis Report

Developed By: Muhammad Taha Imran

Project Background

- Sweet Crust Bakery operates six stores across Lahore, Karachi, Islamabad, and Rawalpindi, sourcing ingredients from six different suppliers.
- Some stores frequently run out of key ingredients while others overstock, and there's no clear way to track which suppliers are actually reliable.
- On top of that, nobody has a clear picture of how much money is being lost to spoiled stock and missed sales due to these stockouts.
- This project analyzes a full year of ordering and stock data across all six stores to pinpoint exactly where these problems are happening and what can be done to fix them.

Data Overview

- The analysis is based on a full year of operational data from 2025, covering daily inventory activity across all six Sweet Crust Bakery stores.
- It tracks 12 core ingredients (including flour, sugar, butter, eggs, milk, and chocolate) organized into four categories: Dry Goods, Dairy, Baking Essentials, and Perishables.
- Five datasets were combined for this analysis: store information, ingredient details, supplier information, daily inventory records showing stock levels, usage, and spoilage, and purchase order records showing what was ordered, from which supplier, and when it actually arrived.



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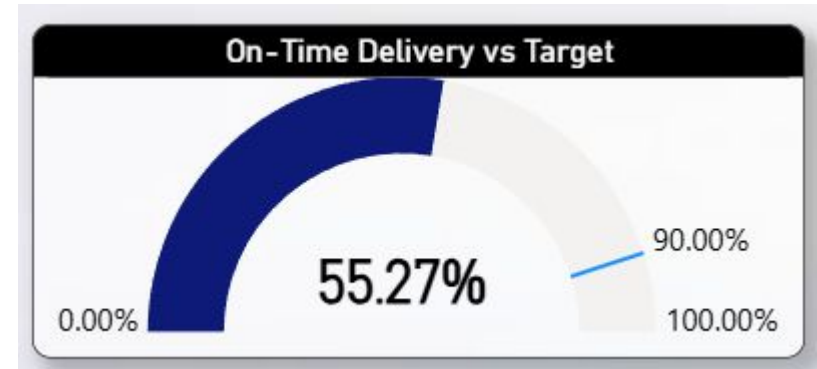
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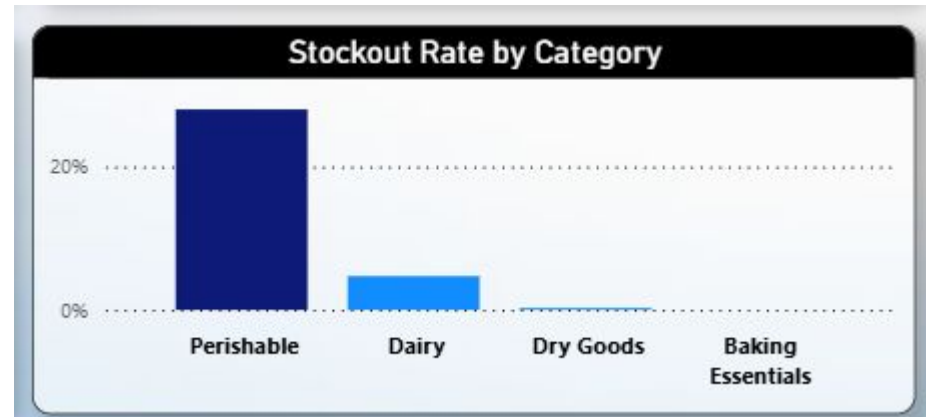
Delivery Performance Falls Well Short of Target

- Only 55.27% of supplier deliveries arrive on time, against a reasonable target of 90%.
- This is not one supplier's problem, every supplier the bakery works with falls short of the target to some degree.
- Late deliveries are the single clearest explanation for why certain ingredients repeatedly run out, since a delayed delivery leaves no buffer stock in reserve.
- This is the first problem worth fixing, because it has knock-on effects on almost every other finding in this report.



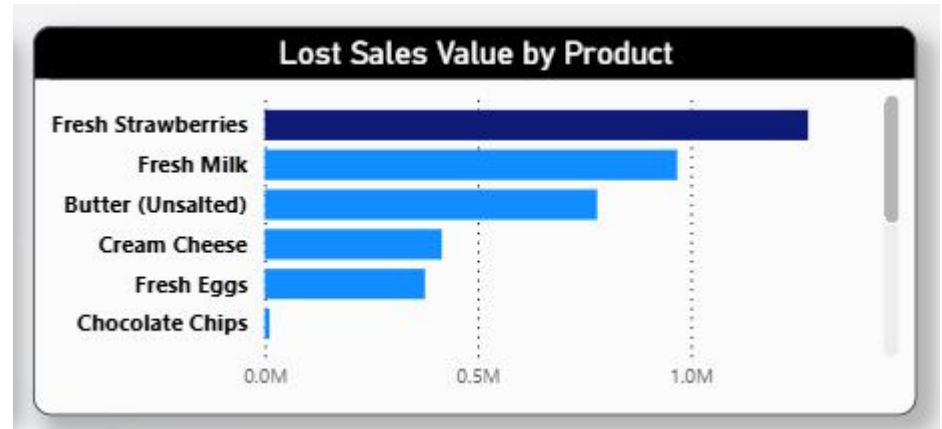
Stockouts Are Concentrated in Perishable Items

- Perishable ingredients run out roughly 28% of the time, far higher than any other category.
- Dairy products stock out about 5% of the time, while Dry Goods and Baking Essentials almost never run short.
- This pattern shows the stockout issue is not about poor ordering across the board, it's specifically about how short-shelf-life items are managed.
- Fixing this means changing how strawberries, milk, butter, and cream cheese are ordered, not overhauling the entire ordering process.



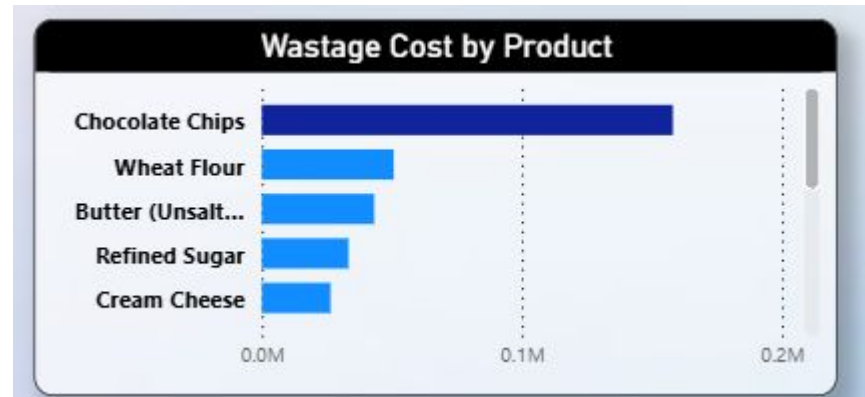
Stockouts Are Costing Real Money

- Fresh Strawberries and Fresh Milk account for the largest share of lost sales value from stockouts.
- Every day an ingredient runs out, that missed usage converts directly into lost sales value, it is not just an operational inconvenience.
- Because lost sales value is concentrated in just a few ingredients, fixing the ordering pattern for those specific items would recover most of this lost value.
- This is a clear, quantifiable case for prioritizing perishable-item ordering improvements first.



Wastage Is Concentrated in a Few Categories

- Dry Goods and Dairy together account for nearly 91% of total wastage cost, more than Perishables alone.
- Chocolate Chips, Wheat Flour, and Butter are the three highest-cost items lost to spoilage and damage over the year.
- This is a smaller cost than stockouts overall (PKR 388K vs. the value of lost sales), but it's a cost that can largely be eliminated through smaller, more frequent ordering.
- Reviewing order sizes for these specific ingredients is a low-effort, high-return fix.



Supplier Reliability Varies Significantly

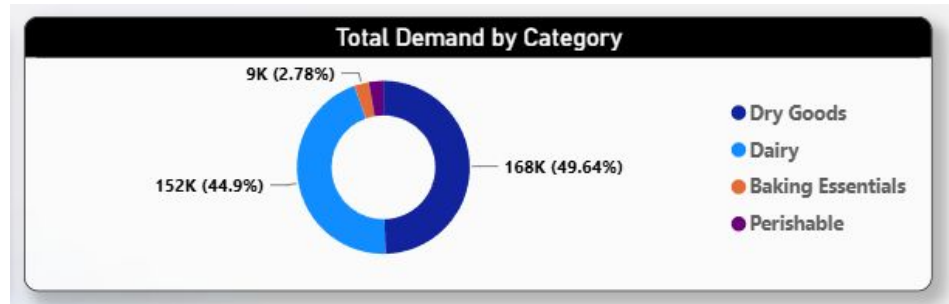
- Al-Rehman Flour Mills is the most dependable supplier, with the highest reliability score (0.92) and fill rate (98.32%).
- Universal Ingredients Ltd has by far the longest average lead time (6.49 days), more than double most other suppliers.
- Fill rates stay strong across all suppliers (97–99%), which means when a delivery is late, it's a timing issue, not a shortage issue.
- This distinction matters: the fix for a slow supplier (renegotiate lead times) is different from the fix for a short-shipping one (renegotiate quantities).

Supplier Performance Summary						
Supplier Name	Reliability Score	Avg Lead Time	On-Time Delivery %	Fill Rate %	Total Spend	
Punjab Dairy Distributors	0.85	2.52	52.93%	97.37%	75917090.1	
Universal Ingredients Ltd	0.83	6.49	54.90%	97.95%	35459377.4	
Al-Rehman Flour Mills	0.92	3.38	54.98%	98.32%	14785773.3	
National Sugar Traders	0.90	4.17	53.91%	98.83%	10834749.7	
Total		2.93	55.27%	97.90%	147561320.1	

Demand Spikes Predictably Around Key Seasons

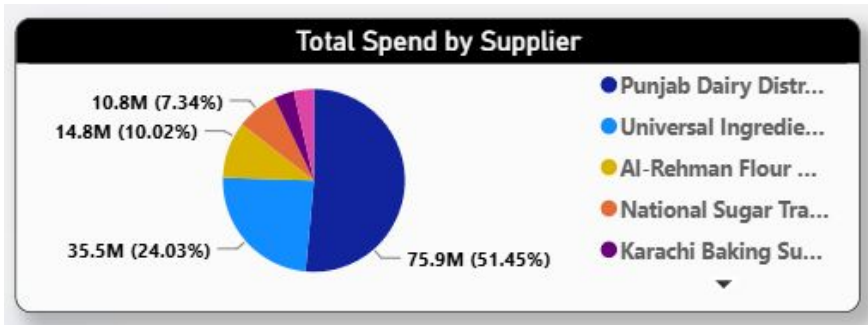
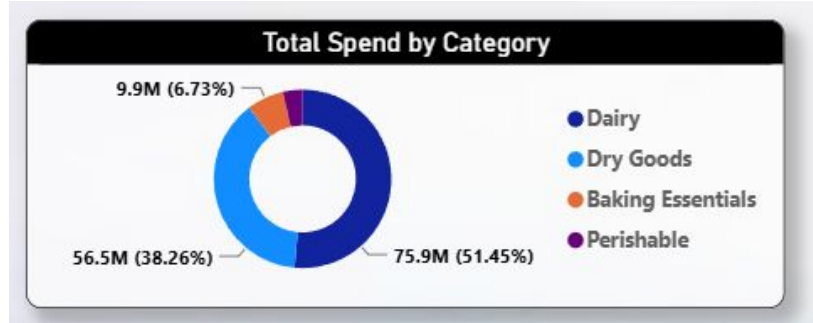


- Wedding season (November–February) drove the single largest seasonal demand increase of the year, totaling nearly 59,600 units.
- Ramadan and Eid-ul-Fitr together pushed demand for sugar, butter, chocolate, and cream cheese up significantly above baseline.
- Because these seasonal patterns repeat every year, they are entirely plannable in advance rather than something the bakery has to react to.
- Building seasonal buffer stock ahead of these known periods would directly reduce stockouts during the bakery's busiest, highest-revenue weeks.



Spend Is Heavily Concentrated in Dairy

- Dairy and Dry Goods together account for nearly 90% of total ingredient spend across the year.
- Punjab Dairy Distributors alone represents 51.45% of total supplier spend, by far the bakery's largest single supplier relationship.
- This level of concentration means the bakery's cost efficiency depends heavily on this one supplier relationship performing well.
- It also means even small improvements in how Dairy ingredients are ordered and managed will have an outsized effect on overall costs.



Recommendations

- Talk to the suppliers who are late most often, and either agree on a more realistic delivery schedule with them, or keep a little extra stock on hand for the items they supply.
- Order strawberries, milk, butter, and cream cheese more often and in smaller amounts, since these are the ingredients running out the most.
- Find a second supplier for strawberries to cover June through October, since the current supplier can't keep up with demand during those months.
- Order less butter, milk, and cream cheese at a time, since these three ingredients are responsible for most of the spoiled and wasted stock.
- Focus first on the Gulberg and Satellite Town stores, since these two locations are running out of stock more often than the rest.

Recommendations

- Stock up on key ingredients a couple of weeks before Ramadan, Eid, and wedding season, since demand for these items rises sharply every year during these times.
- Keep more stock on hand for weekends, since Friday through Sunday are consistently the busiest days of the week.
- Judge suppliers on two separate things: whether they deliver on time, and whether they deliver the full amount ordered, since these are two different problems.
- Remember that stockouts and wastage are currently costing the bakery about PKR 4.39 million a year, so fixing the biggest problems first will save the most money.
- Once the changes work well at Gulberg and Satellite Town, roll them out to the other four stores.